

Teaching Statement

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As an instructor, I aim to create an inclusive atmosphere where students feel seen, supported, and empowered to contribute regardless of gender, race, background, or prior exposure to computing. I hold the perspective that if an teacher is unable to present a concept in a way that is understandable to all students, then there remains more for the instructor to learn. Throughout my career, each course I have taught has continued to deepen my knowledge. I believe that good educators can help students spark their passion for learning and encourage the development of their own ideas.

Teaching Philosophy

My teaching philosophy is grounded in care, clarity, practice, and adaptability, with a strong focus on helping students truly understand the material while feeling supported and confident. I will conclude by outlining my teaching approach through the following key principles:

1. Student-Centered and Adaptive Learning

I approach teaching with the goal of creating a supportive learning environment where students feel comfortable asking questions and never feel “dumb” for what they do not yet understand. From my experience in the classroom, I have found that students learn best when they feel safe to make mistakes, so I emphasize encouragement, patience, and clear explanations to help them build confidence rather than anxiety around difficult concepts. I have worked intentionally to create a classroom where students feel comfortable asking questions, engaging in discussion, and admitting confusion without fear of judgment. I use inclusive language in my teaching to help all students feel comfortable. For example, instead of assuming everyone remembers past material, I invite the class to review it together so no one feels left out or embarrassed. I reach out early to the students who have missing assignments or are not participating in class via a friendly email. Most students often need last-minute help with course materials during exams. I try to keep myself available before a deadline or exam for student questions via email or office hours. One time, one of my students had a leg injury, and could not attend class for several days. I took additional time to help him with all the missing lectures. My students have reported:

- *Friendly during class, responded very well when questions were asked.*
- *She was always willing to answer questions.*
- *Once I came in to office hours she explained what i missed very well.*
- *I was able to go to her office hours and get feedback on the homeworks she assigned which made me do better on the exams.*

I am still at the beginning of my teaching career, and I’ve had the opportunity to work with and learn from many great professors at Baylor University. I am also constantly aspiring to improve my teaching skills and further develop as an educator. I have shaped my teaching style to be more relaxed and conversational during lectures and office hours, because I’ve seen that students are more comfortable asking questions and making mistakes when the environment feels low-pressure and informal. I prioritize adjusting my ways of presenting the same concept to different student needs, and I conduct regular check-ins to support students with different backgrounds and learning styles. Once after the first midterm, I asked students to take a post-exam reflection survey to understand where they were in the course at the beginning of the semester to identify any learning gaps. Another time, after discussing poor grades with students, I decided to provide sample tests before the exam. These are from my students:

- *I think it was valuable and respectful when she would ask us for feedback on the structure of the class and how we should approach the lesson (such as doing more in-class practice problems) and would try to implement our suggestions into her teachings methods.*
- *The instructor was extremely willing to work with students based on exam grades and I really appreciate it. She listened to input from students, and both worked together to do better on the next exams and actually create a solution that worked for everyone. The professor did a great job in adapting to students need. She implemented a review for us which was helpful for test.*
- *I really did enjoy the homeworks this semsters, they helped me while i studied for exams. Also the sample exams were a godsent!*
- *The homeworks prepared us for the test, but it would be helpful to know the format of the test before taking it. The second test we took she gave us a practice test which helped.*

I keep a flexible schedule and adjust my course plans when students need it. I have provided additional time for assignments and makeup exams for valid personal reasons or illness. This helps students stay on track without any unnecessary stress about missing anything. I try to give students enough time to work on materials and clear feedback so they can improve and meet the course goals. I make sure to return feedback on assignments before the next one is due, so students can learn from their mistakes and do better next time. Although this takes careful planning and coordination with teaching assistants, it really helps students grow and succeed in the course. I try to make myself available to students as often as possible in many different settings. In addition to regular office hours, I try to respond via email as soon as possible. Beyond the classroom, I see my role as a mentor, supporting students through office hours, informal discussions, and academic guidance. In one instance, I reached out to a student about her midterm performance and learned that she was not enjoying computer science, which explained her low exam score. We have discussed about changing her major and later came to know that she did change it. This experience reinforced my belief that meaningful conversations can help students better understand their academic goals and make informed, confident decisions about their paths forward.

I focus on delivering adaptive and flexible lectures. During an in-person class, I upload slides and related resources at the beginning of the semester so that anyone who wants to access those can have them before a class, or in case of missing class. During online classes, I implement a flipped classroom approach, where I upload pre-recorded lectures, assign readings before the week and discuss the watched lecture by using more examples. From teaching multiple online classes, I realized that prerecorded lecture videos are very helpful, as a lot of students struggle to keep their focus during an online class. I give them specific instructions on how to be successful taking an online course. These are from students who took the online courses:

- *The video call lectures went in detail over problems that the video lectures may have not explained clearly. It was practically required to attend both and understand the material.*
- *I found the availability of the professor and the reiteration of concepts until complete understanding to be the most useful part of this class. The flipped classroom actually helped to engage learning, and the assignments further allowed me to grasp the concepts.*
- *This class is instructed well and I am impressed with how well this hybrid math class has turned out.*

2. Conceptual Understanding

One of my core teaching philosophies is that ensuring every student truly understands the material is the most important responsibility of a teacher. I try to make the subject accessible by providing clear explanations, using real-world examples, and adapting my teaching to different learning styles. I revisit difficult concepts until they are fully understood. When teaching, I like to give students insight into my own thought process and how I approach a problem. In an one-to-one setting, I reverse this approach and ask the student to describe their thought process as we work through the topic together. When teaching online, I use an iPad to show prepared lecture slides and include blank slides every few pages. These blank slides let me write notes, answer student questions on the spot, and adjust the lecture as needed. During an in-person class, I heavily use the board to write and explain everything. One of the students said,

I found I learned and understood the subject matter better when Professor Tayba lectured without a PowerPoint and wrote her notes on the board. She would write theorems and work out examples step by step. This really helped me understand what I was doing on paper, not just memorizing the steps to get the problem right.

I use and show a ton of examples from a topic to make sure students get a whole picture from all perspectives. It helps them connect abstract ideas to real examples, making the material easier to understand. From my students in Discrete Structures class, when asked what was the most valuable aspect of the course,

- *I found that class time examples were the most valuable.*
- *Yes, she showed a lot of examples and explained each step in detail.*
- *The numerous examples we worked in class and in the homework, especially when dealing with permutations and combinations, which were particularly confusing for me.*
- *The most valuable aspect of this course was the homework and the practice problems that we did during class. The questions were comprehensive and the examples made it easy to understand the lecture and study for exams.*
- *going through examples and actively making the class practice techniques we learned about in class, also, the homework assignments definitely helped prepare me for the test and were where the majority of my comprehension happened.*
- *The examples we go over in class are very helpful, they clearly showcase how to approach different types of questions.*

I also give short in-class quizzes based on the most recently covered topic to reinforce learning and check student understanding. Multiple students reported that they benefited greatly from this approach. In courses that covered multiple topics, I assigned daily homework based on a single class session. Although some students did not initially find this assignment-heavy approach helpful, many later realized that it significantly supported their final exam preparation by allowing them to review graded assessments. From my students,

- *I really liked the quizzes that she provided, because it was a good review of what we had just learned and it helped explain some concepts that I didn't understand, as well as the homework.*
- *The homework we were given was the most helpful to learning the subject as practice was needed for many of the concepts we were learning.*
- *I feel like the heavy weight on homework allows for practicing the topics to be more important then just a test and creates a very stress free environment.*
- *The homeworks for each topic was really helpful for making sure I understood the material.*
- *The homework assignments were helpful to practice the concepts that we learned in class. She had a homework for each topic and idea that we covered to make sure we understood the material.*
- *I found the wide access to relevant study materials and practice from the homework assignments to be the most valuable.*
- *I liked the structure of the course and how the homework assignments were laid out, they helped reinforce my learning of the subjects. I also found it helpful when she posted the powerpoints because it was very nice to be able to go back and reference them.*

3. Engagement and enjoyable learning

I believe effective teaching is based on creating an engaging classroom environment and remaining adaptable to students' needs. I structure my lectures to create an atmosphere that supports activities like pair work and group discussion. Due to different personality traits (shyness), some students prefer not to ask questions or actively participate in such activities. If someone is not participating in discussion, I make sure to individually check up on that student to ask if they need any help or have any questions. For group projects, I realized that asking students to confidentially evaluate their team members helps identify capabilities and areas needing improvement, which encourages self-awareness and growth. It also resolves issues and builds better collaboration skills. For my hybrid course, I assign readings beforehand and encourage

questions and comments during class time to help students stay involved and connected to the material. I intentionally use creative moments, such as bringing small treats/objects to illustrate a concept, to keep students actively engaged in the learning process. I try to excite students' interest by using examples that are tied to familiar topics and popular culture. For example, I referenced a TV series, "Money Heist," to explain recursive algorithm steps. I've demonstrated the Monty Hall Puzzle by referencing an episode from a famous TV series, "Brooklyn 99." I've found that this is a valuable tool in keeping students' attention. I also include movies and TV series in assignment questions to keep students engaged and make the work more interesting rather than boring. My teaching style can be best described by the following quote from students:

- *For one of the questions we were going over she was engaging with students by bringing us chocolate to relate it to the question.*
- *Thank you for being a fun professor to learn from!*
- *She's a very nice professor and always trying to engage with students in a joyful manner.*

I believe that all students benefit from learning in an environment that is diverse in background, identity, and experience. My own journey in computer science has reinforced this belief. I was the among the very few female student in the room during my undergrad and graduate journey. That experience made me acutely aware of how isolating it can feel to be the "among very few," but it also motivated me to create classrooms where no student feels out of place. This is a email I had after my first semester at Baylor teaching:

"I just wanted to thank you for being my professor this year! You were a great teacher for a hard subject and I really appreciate how much help and how understanding you've been, especially with the fact that you were teaching so many other classes. . . . Also, it's a small thing, as I don't mind being the only girl in the class, but I was still really glad to have a female professor who made me feel not so alone. . . ."

4. Learning Through Practice

I firmly believe that people can get better at things over time through consistent practice and proper application of concepts. My courses are structured around regular homework assignments for each topic, short formative assessments, and programming assignments that allow students to apply theoretical ideas in practical contexts. I try to create a lower-stress learning environment that prioritizes conceptual understanding and long-term retention by placing more emphasis on practice/assessments rather than testing alone. My teaching process emphasizes well-balanced homework assignments that are focused on essential concepts, and scheduled in a way that supports steady learning without overwhelming students. From students,

- *I feel like the heavy weight on homework allows for practicing the topics to be more important then just a test and creates a very stress free environment.*
- *The homeworks are not overly lengthy, and are the perfect amount of questions for what is needed to know for that section, and the spacing of when they are due is good.*
- *found the most valuable aspect of this course to be the pacing. We were able to actually learn the material in class without feeling rushed, and the professor explained everything really well.*
- *The programming assignments and actually going through examples in class were the most valuable aspects of this course, I could see what we were talking about.*
- *She had programming assignments that made us take a the laws that we learned and apply them in our code.*
- *Frequent examples of concepts talked about was very helpful.*

Teaching Experience

My teaching experience began years ago with tutoring three to four different students each week as a means of supporting myself financially during high school. Since then, I have always viewed teaching as an integral

component of my own learning.

Previously, I worked as a lecturer for two years in Bangladesh during 2018-2020, teaching full-time and managing complete courses independently in large class sizes (70–90 students). Across these roles, I have taught Data Structures, Digital Logic Design, Database Management Systems, Computer Networks, Advanced Java, System Analysis and Design, and Numerical Analysis. I designed course content from scratch and modified existing materials, including developing and tailoring lectures and assessments. I mentored students in research and classroom projects. I supervised two undergraduate students in their final year projects. Other roles include participating in academic committees, curriculum development, and administrative duties related to the academic department.

I was hired as an adjunct lecturer at Baylor University in 2023 to teach CSI 2334: Introduction to Computer Systems, which focuses on the assembly language and computer architecture. I was the instructor of record for two sections with 45 students, responsible for lecturing, writing and grading exams, and coordinating with the teaching assistant. Despite the compressed schedule and the fact that it was my first experience as the primary instructor in the U.S., it went well. I have taught CSI 2350: Discrete Structures for four semesters in the past years in multiple sections, focusing on helping students build strong foundations in logic, proofs, sets, graphs, algorithms, and combinatorics. During this time, I have worked with classes ranging from small sections of about 12 students to medium sections of up to 35 students. My responsibilities include preparing lecture materials for delivery, assignments, and other instructional content; designing/updating course content; helping students; preparing and grading exams and coordinating with teaching assistants.

In Spring 2021, I was TA for CSI 4330: Foundations of Computing, in which I managed grading, wrote homework solutions and ran discussion sessions for students. In Fall 2021, I was TA for CSI 2350: Discrete Structures, where solving and grading assignments were my main responsibilities. From spring 2022 to Spring 2023, I was TA for CSI 1430: Intro to CSI I with Lab where I held the lab sessions for all the sections. I also held weekly tutoring sessions for students for this course who needed additional help outside of classroom with their programming assignments.

I have attended several teaching-related workshops organized by the Academy for Teaching and Learning at Baylor University. These workshops covered first-day practices, classroom culture, course design, assessment strategies, and serving diverse students. I also received guidance on understanding students and writing effective learning objectives. In addition, I participated in training on enhancing classroom equity and student learning through transparent teaching methods. I have attended a seminar on how to learning from students and what they teach us about effective instruction.

Teaching Interest

My interest lies in the following areas:

- Theoretical courses: Discrete Structures, System Analysis and Design.
- Core introductory courses: Introduction to Computer Programming, Data Structures, Algorithms, and Database management courses.

Course Evaluation

My course evaluations show consistent improvement each semester, reflecting my growing understanding of the material and my increasing ability to explain concepts effectively to students. The following are selected quotes from anonymous course evaluation from Fall 2023 to Fall 2025 at Baylor University:

- She has an incredible future being a professor and was one of the most kind, compassionate people I've ever met. She works hard despite a busy schedule and goes above and beyond for us

student.

- This course provided me with good foundations in data science structures which I couldn't have gotten anywhere else.

- The professor is really nice, and I can tell she cares about the transfer of knowledge to the students.

- She was a very caring professor, and did her best to make sure we had the resources we needed to learn.

- When actually having one on one conversations with professor Tayba, I found her to be a really nice and pleasant person. I think her personality and willingness to help students even on weekends is what keeps this class from being a complete waste of time. She is a wonderful person and I really do appreciate the time she takes to help us be successful."

(about the comment "complete waste of time"; not all the students were happy to be in a course where it teaches how to write assembly language code.)

Summary

As I have grown as an educator, I have become more reflective and adaptive; using student feedback to refine my pacing, examples, and assessments so learning feels supportive rather than overwhelming. I hope to continue developing more engaging, student-focused activities both inside and outside the classroom that strengthen understanding, build confidence, and help students leave my courses feeling capable rather than discouraged.